

JKBOSE 2026

CLASS 11TH

CHEMICAL BONDING AND MOLECULAR STRUCTURE

LECTURE-2

CHEMISTRY

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ON
TOP



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Topics To Covered

XUV.

- 1 Covalent Bond
- 2 Writing Lewis Structures or Electron Dot Structures ✓
- 3 ~~Limitations of Octet Rule~~ X

⋮

REVISION.

① Why do atoms combine.

a) Octet Rule.

b) To acquire minimum energy.

② Lewis dot Symbols



③ IONIC BOND.



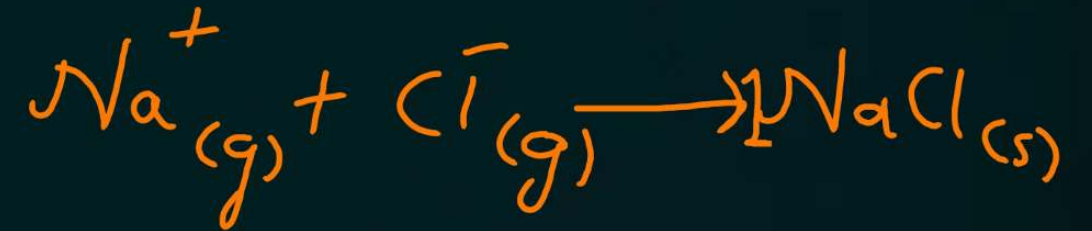
Noble gas configuration.

Factors affecting formations of ionic bond $\rightarrow$

- 1) Low Ionisation energy $\rightarrow$ One atom should have low ionisation energy, lowest the ionisation energy more easily will be the ionic bond formation. e.g. Group 1 & Group 2 elements have least Ionisation energy.
- 2) High $\bar{e}$ gain enthalpy $\rightarrow$ Other atom should have more -ve $\bar{e}$ gain enthalpy, more -ve the $\bar{e}$ gain enthalpy more easily will be the ionic bond formation. e.g. Group 16 & 17 have highly -ve $\bar{e}$ gain enthalpy.

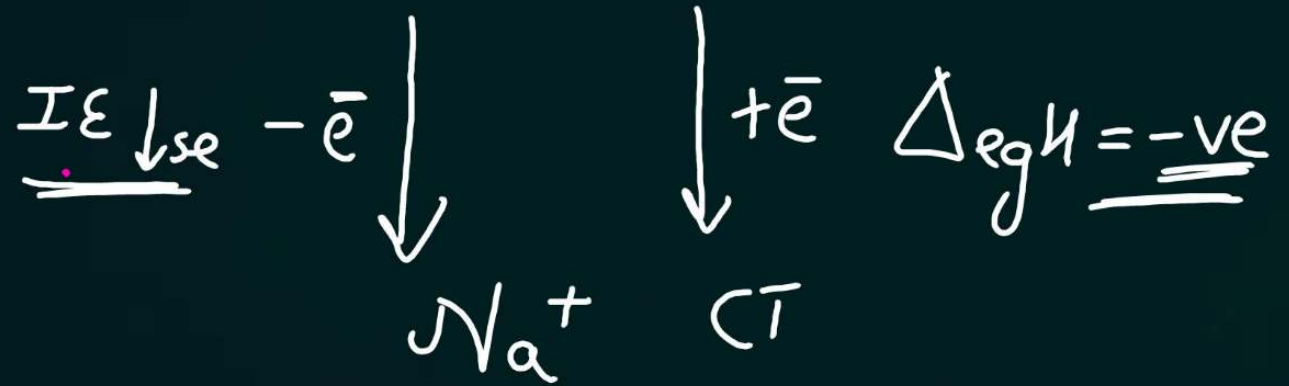
③ HIGH LATTICE ENERGY.

Energy released when two oppositely charged gaseous ions combine to produce 1 mole of solid ionic compound. is called Lattice energy. e.g. formation of NaCl.

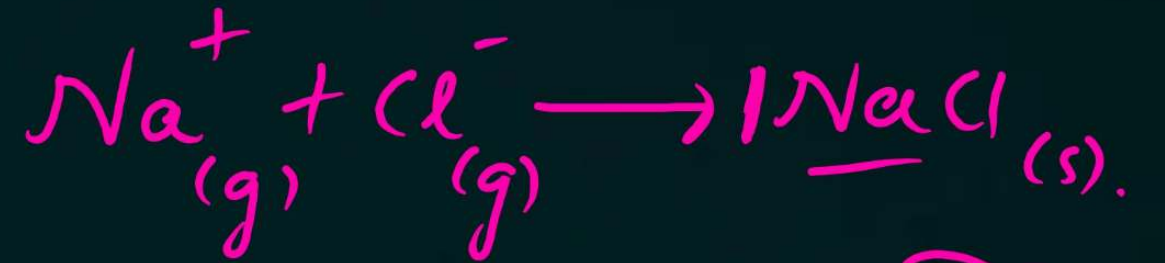


High the lattice energy more easily will the formation of ionic compound. Denoted by 'U'.

Na Cl.



Energy Release 'U' ↑ se.



= U ↑

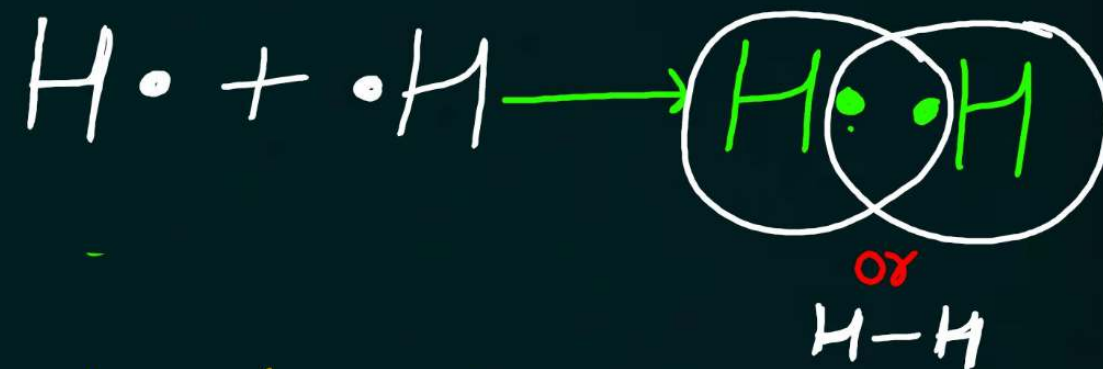
Covalent Bond

Bond formed by the mutual sharing of electrons between two same or different elements is called covalent bond. Covalent bond is formed between non-metal.

No. of electrons shared by an atom in a covalent bond formation is called its covalency and the compounds having covalent bonds are called covalent compounds.



- If two atoms share one pair of e^- in a covalent bond, the single covalent bond. e.g. in H_2 .

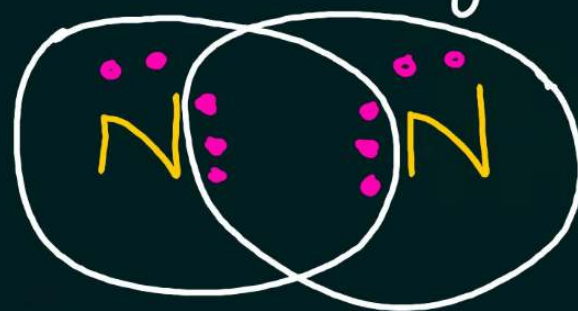
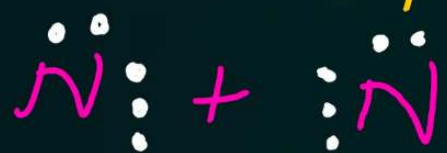


- If two atoms share two pairs of e^- in covalent bond, then double covalent bond is formed. e.g. in O_2 .

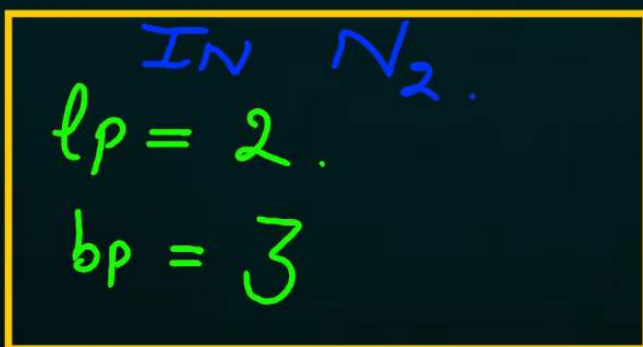


e pair present in bond are called bond pair of electron & others are called lone pair of electron.

- If two atoms share 3 pairs of e in a covalent bond, then triple bond is formed. e.g. in N_2 .



OR



Covalent Bond



LEWIS DOT STRUCTURE $\Rightarrow$

OR

ELECTRON DOT STRUCTURE.

i) Total valence $\bar{e} = \text{Valence } \bar{e} \text{ of each atom} + (-ve \text{ charge}) - (+ve \text{ charge})$

ii) Selection of central atom.

a) Atom present in least number.

b) Least electronegative atom

c) H & F never act as central atom.

iii) Total no. of bonds formed by central atom.

$$= \frac{\text{Required } \bar{e} - \text{Total } \bar{e}}{2}$$

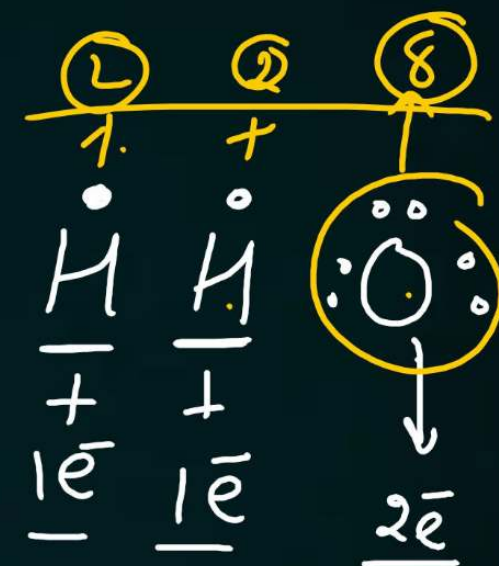
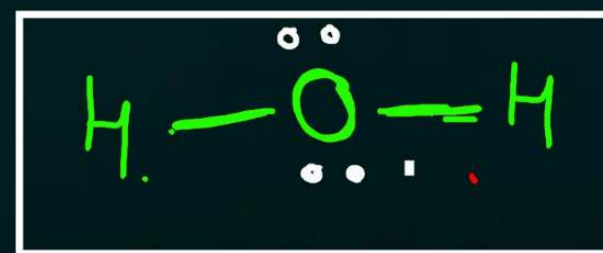
Q. Draw the $\bar{e}$ dot str. of H_2O , NH_3 , CH_4 etc.

i) H_2O

$$TVE = 2 + 6 = 8$$

Central atom = O

$$\text{Total bond} \Rightarrow \frac{\text{Reqd } \bar{e} - \text{Total } \bar{e}}{2} = \frac{12 - 8}{2} = \frac{4}{2} = 2$$

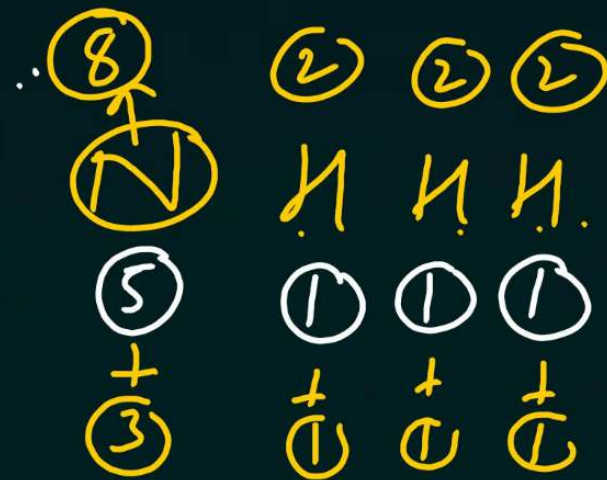


b) NH_3 .

$\text{TV}\bar{e} \rightarrow 5 + 3 = 8\bar{e}$

Central atom $\rightarrow \text{N}$

Total bonds $\rightarrow \frac{\text{Reqd. } \bar{e} - \text{Total } \bar{e}}{2} =$
 $= \frac{14 - 8}{2} = \frac{6}{2} = 3$



iii). CH₄

8 2 2 2 2
C H H H H.

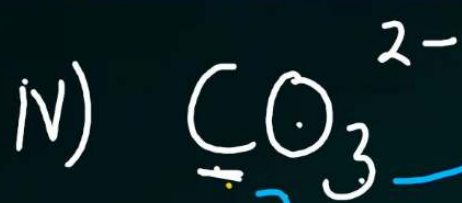
$$\text{TV}\Sigma \rightarrow 4 + 4 = 8e^-$$

Central atom. = C

$$\begin{aligned} \text{Total bond.} & \rightarrow \frac{\text{Reqd. } e^- - \text{Total } e^-}{2} \\ & = \frac{16 - 8}{2} = \frac{8}{2} = 4 \end{aligned}$$



Covalent Bond

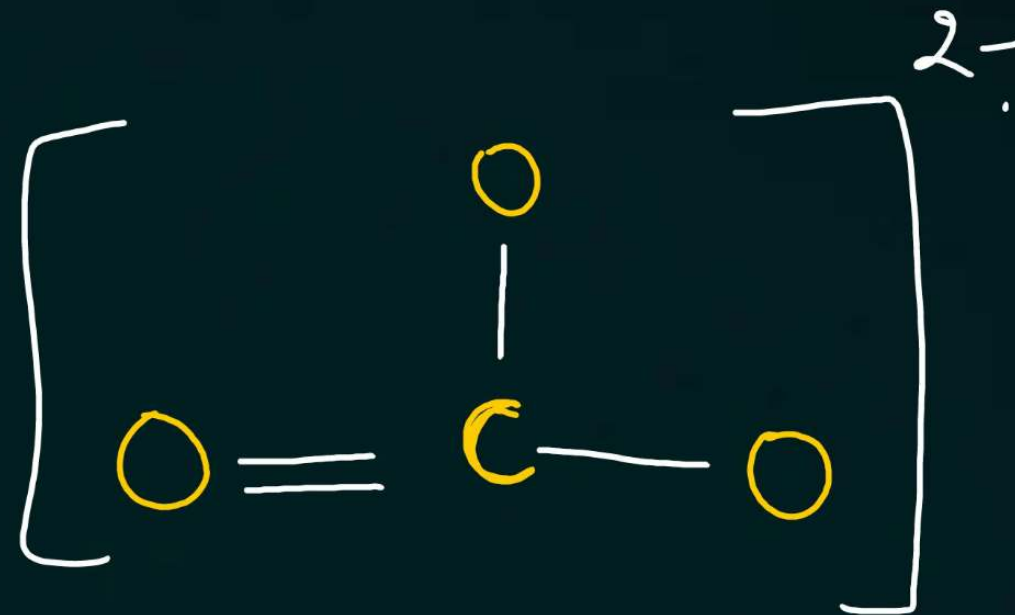


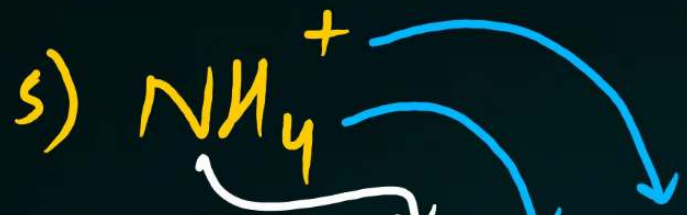
$$\text{TV} \varepsilon = 4 + 3(6) + 2\bar{e} = 24\bar{e}$$

Central atom $\rightarrow \text{C}$

$$\text{Total no. of bonds} = \frac{\text{Reqd. } \bar{e} - \text{Total } \bar{e}}{2}$$

$$= \frac{32 - 24}{2} = \frac{8}{2} = 4$$

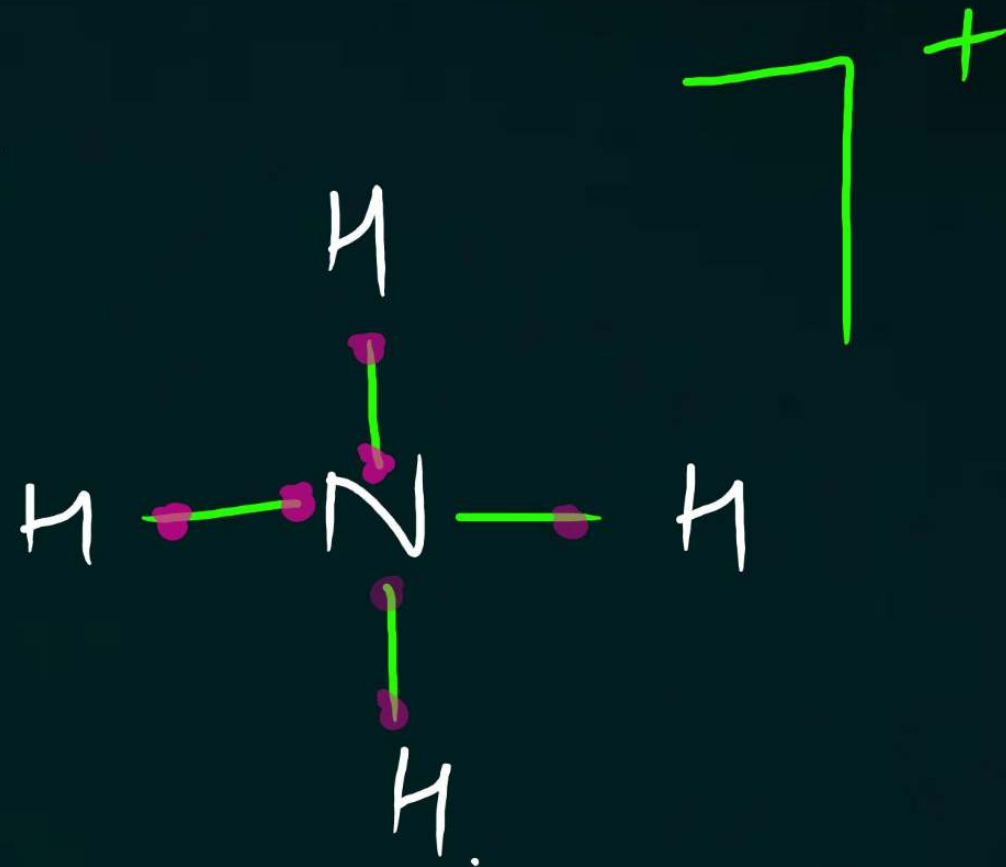




$$\text{TVE} = 5 + 4 - 1 = 8$$

Central atom = N

$$\begin{aligned} \text{Total no. of bond} &= \frac{\text{Reqd. } \bar{e} - \text{Total } \bar{e}}{2} \\ &= \frac{16\bar{e} - 8}{2} = \frac{8}{2} = 4. \end{aligned}$$



Reqd e

$$\text{CH}_4 = 8 + 8 = 16e^-$$

$$\text{CO}_2 = 8 + 16 = 24e^-$$

$$\text{NH}_3 = 8 + 6 = 14e^-$$

Reqd e in

$$\underline{\text{O}}_3 = 24e^-$$

$$\text{H} = 2e^-$$

Writing Lewis Structures or Electron Dot Structures

EXAMPLE:



HOMEWORK.



Lewis DOT STRUCTURE.

Thank
You